Tony Chen

tonychengangyi@gmail.com | github.com/tonychen-2006 | tony-chen.dev | +1 (778) 814-6602

EDUCATION

The University of British Columbia — Vancouver, BC
Bachelor of Applied Science in Computer Engineering — Expected May 2028
Relevant Coursework: Software Construction, Computing Systems, Algorithms and Data Structures, Circuit Analysis

TECHNICAL SKILLS

Languages: C, C++, Python, SystemVerilog, ARM Assembly, MATLAB/Simulink
Embedded Systems: STM32, ESP32, FreeRTOS, Zephyr RTOS, device drivers, HALs, interrupts, DMA, CMake, CAN, UART, I2C, SPI
Debugging and Tools: GDB, J-Link, logic analyzer, oscilloscope, Git, GitHub Actions, Make, KiCad

WORK EXPERIENCE

Verdi — Vancouver, BC
Firmware Developer Intern — *Jan 2026 – Apr 2026*

- Led migration of an embedded C codebase from FreeRTOS to Zephyr RTOS, re-architecting task scheduling and driver interfaces and adapting CMake builds and device-tree overlays for multi-board support.
- Implemented real-time micro-valve control using GPIO and RTOS threads, with safe boot, continuous state recovery, and watchdog-supervised operation.
- Integrated a LoRa radio module, implementing hardware initialization, packet handling, and RTOS-safe asynchronous message processing.
- Built GitHub Actions CI pipeline for board-specific builds, unit tests, and automated clang-format on pull requests.

EIM Technologies — Richmond, BC
Embedded Systems Engineer Intern — *May 2026 – Aug 2026*

- Engineered an educational audio DSP platform, prototyping the analog circuitry and writing STM32 firmware to demonstrate AUX/ADC acquisition, FFT spectrum analysis, and FIR/IIR digital filtering.
- Developed an FPGA–STM32 processing system, partitioning high-speed sampling, triggering, timestamping, and parallel preprocessing onto FPGA logic while an STM32 manages configuration and data transfer over SPI and custom hardware prototype in KiCad

UBC Solar Racing — Vancouver, BC
Senior Embedded Systems Designer — *Sep 2024 – Present*

- Led firmware development for the Motor Control subsystem, implementing vehicle drive-state (drive/reverse/park) control as a safety-critical state machine using ADC-based potentiometer input, CAN handling, and motor-controller actuation through I^2C-controlled DAC and GPIO control.
- Built an STM32 bootloader for remote firmware updates over CAN, with CRC validation, resumable flashing, and safe application handoff.
- Designed a FreeRTOS-based cruise control system, using a PI controller with real-time force modeling (drag, rolling resistance, hill climbing) for speed regulation.
- Developed STM32 validation infrastructure with Ceedling unit tests, Python-based CAN/state simulation, and hardware test firmware to validate PCB GPIO and serial communication interfaces.
- Developed Telemetry Board firmware for cellular/radio communication, bridging vehicle CAN data over UART with message filtering and rate limiting for bandwidth-efficient transmission.
- Built a Raspberry Pi/Python telemetry gateway to process and stream vehicle CAN telemetry over Ethernet/IP to InfluxDB/Grafana with under 2-second end-to-end latency.
- Implemented GPS data acquisition firmware as a FreeRTOS task that parsed NMEA sentences (GGA, GLL, GSA, RMC, VTG, GSV), validated checksums, and packaged results into CAN frames.

PROJECTS

Custom RTOS

- Built a preemptive bare-metal RTOS for STM32 Cortex-M, implementing task stack initialization, priority-based scheduling, and context switching through SysTick and PendSV exception handlers.
- Implemented semaphores and mutexes with scheduler-integrated task blocking and wakeup.
- Developed low-level peripheral drivers and diagnostics for logging and fault handling..